Personas: One Approved, One Denied

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- Candidate applicants
- Why the model decided what it did
- Age and gender in these explanations
- Summary for the slides

Purpose. This notebook explains *why* the model scores an applicant the way it does, using two real, anonymized holdout applicants: one approved and one denied. Each prediction is split into contributions (SHAP values) that show which parts of the application pushed the predicted risk up or down. Any names used for the personas on the presentation slides are made up.

These two applicants are intentionally selected illustrative examples, not a representative sample. I filtered the holdout to candidates who fit a clear story (including on their *actual* outcome) and then picked one of each by hand. They show how the model reasons about individual cases; they say nothing about how often the model is right or how typical these drivers are.

Model. The scores come from the same refit of the final XGBoost model used in `01_business_impact.Rmd` (holdout AUC 0.7550; the original run scored 0.7556).

Association, not causation. A SHAP contribution shows how much a feature moved *this model's* score, given everything else in the application. It does not show that the characteristic causes default, or that changing it would change the applicant's real risk.

Decision rule used for the personas: the riskiest 10% of applicants by model score are denied; everyone else is approved. Change `deny_share` to try another rule.

```
deny_share <- 0.10
cutoff <- quantile(holdout_prob, 1 - deny_share)

ho <- holdout_data |>
  mutate(row_id = row_number(),
         prob = holdout_prob,
         risk_percentile = percent_rank(prob),
         risk_rank = min_rank(desc(prob)),   # 1 = riskiest applicant in the holdout
         decision = if_else(prob >= cutoff, "Deny", "Approve"))

tibble(`Deny if predicted risk is at least` = pct(cutoff),
       `Average predicted risk` = pct(mean(ho$prob))) |>
  knitr::kable()
```

Deny if predicted risk is at least	Average predicted risk
17.9%	8.0%

Candidate applicants

The filters below were chosen to produce clear, easy-to-explain examples. Because they use the applicant's actual outcome, the candidates are a hand-picked subset by design.

Approved candidate: no credit bureau record (the borrower a traditional bank turns away), repaid, in the safest half of applicants, has at least two outside scores, aged 22 to 35, currently employed.

Denied candidate: in the riskiest 10%, actually had payment difficulties, aged 22 to 45, and does have a credit bureau record (so the denial is not about missing history).

```
profile_cols <- c("row_id", "prob", "risk_percentile", "TARGET", "age_years",
                  "CODE_GENDER", "years_employed", "NAME_INCOME_TYPE",
                  "OCCUPATION_TYPE", "NAME_EDUCATION_TYPE", "NAME_FAMILY_STATUS",
                  "CNT_CHILDREN", "NAME_HOUSING_TYPE", "AMT_INCOME_TOTAL",
                  "AMT_CREDIT", "AMT_GOODS_PRICE", "AMT_ANNUITY", "ltv",
                  "ext_source_mean", "ext_source_count", "has_bureau_record",
                  "bureau_credits", "fields_missing", "NAME_CONTRACT_TYPE")

approved_candidates <- ho |>
  filter(has_bureau_record == 0, TARGET == 0, risk_percentile <= 0.5,
         ext_source_count >= 2, between(age_years, 22, 35),
         not_currently_employed == 0, OCCUPATION_TYPE != "Missing") |>
  arrange(prob) |>
  slice(seq(1, n(), length.out = min(5, n()))) |> round())   # spread across the group

denied_candidates <- ho |>
  filter(decision == "Deny", TARGET == 1, between(age_years, 22, 45),
         has_bureau_record == 1, OCCUPATION_TYPE != "Missing") |>
  arrange(desc(prob)) |>
  slice(seq(1, n(), length.out = min(5, n()))) |> round())

approved_candidates |> select(any_of(profile_cols)) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  knitr::kable(caption = "Approved candidates")
```

Approved candidates

row_id	prob	risk_percentile	TARGET	age_years	CODE_GENDER	years_employed	NAME_INCOME_TYPE	OCCUPATION_TYPE	NAME_EDUCATION_TYPE	NAME_FAMILY_STATUS	CNT_CHILDREN	NAME_HOUSING_TYPE	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	AMT_ANNUITY	ltv	ext_source_mean	ext_source_count	has_bureau_record	bureau_credits	fields_missing	NAME_CONTRACT_TYPE
26352	0.009	0.002	0	33.629	F	10.152	Commercial associate	Sales staff	Higher education	Single / not married	0	House / apartment	360000	630000	630000	63000.0	1.000	0.770	3	0	0	1	Revolving loans
10604	0.028	0.205	0	29.246	F	10.267	Commercial associate	Laborers	Secondary / secondary special	Married	1	House / apartment	67500	152820	135000	15241.5	1.132	0.552	2	0	0	8	Cash loans
46906	0.036	0.315	0	29.128	F	4.266	State servant	Core staff	Higher education	Married	0	House / apartment	112500	540000	540000	22878.0	1.000	0.533	2	0	0	8	Cash loans
61302	0.045	0.421	0	26.880	F	6.716	Working	High skill tech staff	Higher education	Married	2	House / apartment	450000	258768	216000	23863.5	1.198	0.502	2	0	0	8	Cash loans

row_id	prob	risk_percentile	TARGET	age_years	CODE_GENDER	years_employed	NAME_INCOME_TYPE	OCCUPATION_TYPE	NAME_EDUCATION_TYPE	NAME_FAMILY_STATUS	CNT_CHILDREN	NAME_HOUSING_TYPE	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	AMT_ANNUITY	ltv	ext_source_mean	ext_source_count	has_bureau_record	bureau_credits	fields_missing	NAME_CONTRACT_TYPE
6778	0.053	0.497	0	28.000	F	0.364	State servant	Medicine staff	Secondary / secondary special	Married	1	House / apartment	90000	180000	180000	9000.0	1.000	0.507	2	0	0	8	Revolving loans

```
denied_candidates |> select(any_of(profile_cols)) |>
  mutate(across(where(is.numeric), ~ round(.x, 3))) |>
  knitr::kable(caption = "Denied candidates")
```

Denied candidates

row_id	prob	risk_percentile	TARGET	age_years	CODE_GENDER	years_employed	NAME_INCOME_TYPE	OCCUPATION_TYPE	NAME_EDUCATION_TYPE	NAME_FAMILY_STATUS	CNT_CHILDREN	NAME_HOUSING_TYPE	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_GOODS_PRICE	AMT_ANNUITY	ltv	ext_source_mean	ext_source_count	has_bureau_record	bureau_credits	fields_missing	NAME_CONTRACT_TYPE
22892	0.607	1.000	1	39.329	M	2.261	Working	Drivers	Secondary / secondary special	Civil marriage	1	House / apartment	202500	640080	450000	31261.5	1.422	0.096	2	1	3	49	Cash loans
24700	0.329	0.986	1	28.468	F	1.960	Working	Core staff	Higher education	Married	1	House / apartment	103500	450000	450000	27324.0	1.000	0.252	2	1	17	49	Cash loans
38886	0.261	0.965	1	41.164	M	1.344	Working	Laborers	Secondary / secondary special	Married	2	House / apartment	157500	1303812	1138500	42187.5	1.145	0.261	3	1	9	48	Cash loans
36133	0.215	0.935	1	39.225	M	0.337	Working	Low-skill Laborers	Secondary / secondary special	Civil marriage	0	House / apartment	90000	269550	225000	14620.5	1.198	0.367	3	1	1	48	Cash loans
10930	0.179	0.900	1	38.716	M	1.133	Working	Drivers	Secondary / secondary special	Married	0	House / apartment	157500	835605	697500	24561.0	1.198	0.394	2	1	1	49	Cash loans

Pick one of each. The defaults take the first row of each table; change the numbers to use a different candidate.

```
approved_pick <- 3
denied_pick   <- 1

personas <- bind_rows(
  approved_candidates |> slice(approved_pick) |> mutate(persona = "Approved"),
  denied_candidates   |> slice(denied_pick)   |> mutate(persona = "Denied")
)
```

Why the model decided what it did

```
ex <- explain_applicants(final_internal_xgb, personas)
stopifnot(max(abs(ex$prob - personas$prob)) < 1e-4)  # contributions add up to the score

contrib <- ex$long |>
  mutate(persona = personas$persona[row], theme = theme_of(variable))

by_theme <- contrib |>
  group_by(persona, theme) |>
  summarise(contribution = sum(contribution), .groups = "drop") |>
  filter(abs(contribution) > 0.005) |>
  arrange(persona, desc(abs(contribution)))

by_feature <- contrib |>
  filter(contribution != 0) |>
  group_by(persona) |>
  slice_max(abs(contribution), n = 8) |>
  ungroup() |>
  mutate(direction = if_else(contribution > 0, "raises risk", "lowers risk"))

by_theme |>
  mutate(direction = if_else(contribution > 0, "raises risk", "lowers risk"),
         contribution = round(contribution, 3)) |>
  knitr::kable(caption = "Contribution by theme (log-odds; positive = riskier)")
```

Contribution by theme (log-odds; positive = riskier)

persona	theme	contribution	direction
Approved	Outside credit scores	-0.303	lowers risk
Approved	Education	-0.195	lowers risk
Approved	Loan vs. price of goods	-0.100	lowers risk
Approved	Income	-0.079	lowers risk
Approved	Loan size and payments	-0.078	lowers risk
Approved	Gender	-0.077	lowers risk
Approved	Housing and location	-0.051	lowers risk
Approved	Other application details	0.045	raises risk
Approved	Credit bureau history	0.026	raises risk
Approved	Age	0.023	raises risk
Approved	Family	-0.021	lowers risk

persona	theme	contribution	direction
Approved	Missing information	-0.015	lowers risk
Approved	Job stability	-0.011	lowers risk
Denied	Outside credit scores	1.623	raises risk
Denied	Loan vs. price of goods	0.351	raises risk
Denied	Loan size and payments	0.277	raises risk
Denied	Gender	0.131	raises risk
Denied	Job stability	0.119	raises risk
Denied	Other application details	0.092	raises risk
Denied	Age	0.073	raises risk
Denied	Credit bureau history	0.053	raises risk
Denied	Education	0.048	raises risk
Denied	Missing information	0.039	raises risk
Denied	Income	0.034	raises risk
Denied	Family	0.029	raises risk
Denied	Housing and location	0.009	raises risk

```
by_feature |>
  select(persona, column, value, contribution, direction) |>
  mutate(across(c(value, contribution), ~ round(.x, 3))) |>
  knitr::kable(caption = "Top 8 individual drivers per persona")
```

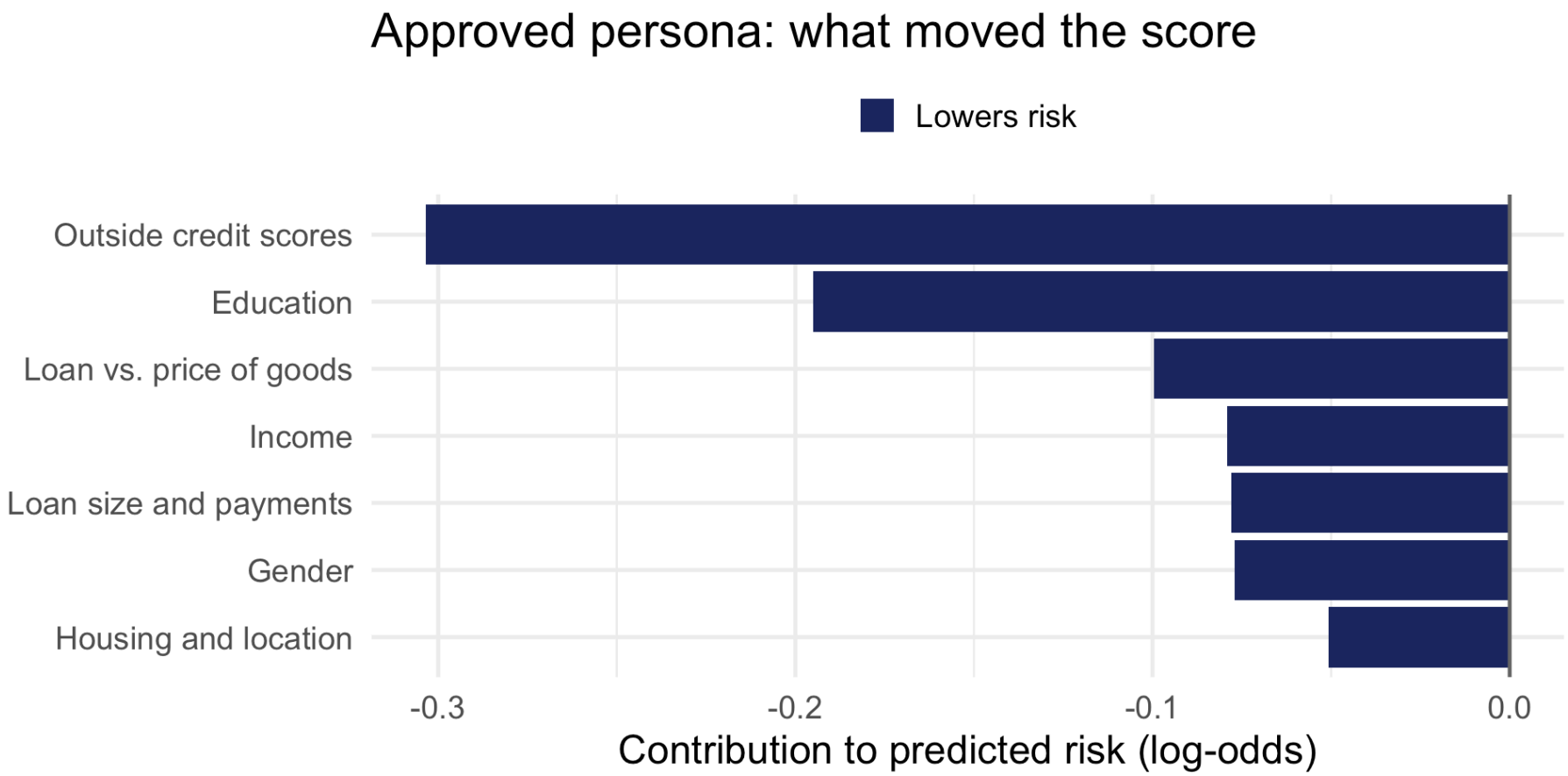
Top 8 individual drivers per persona

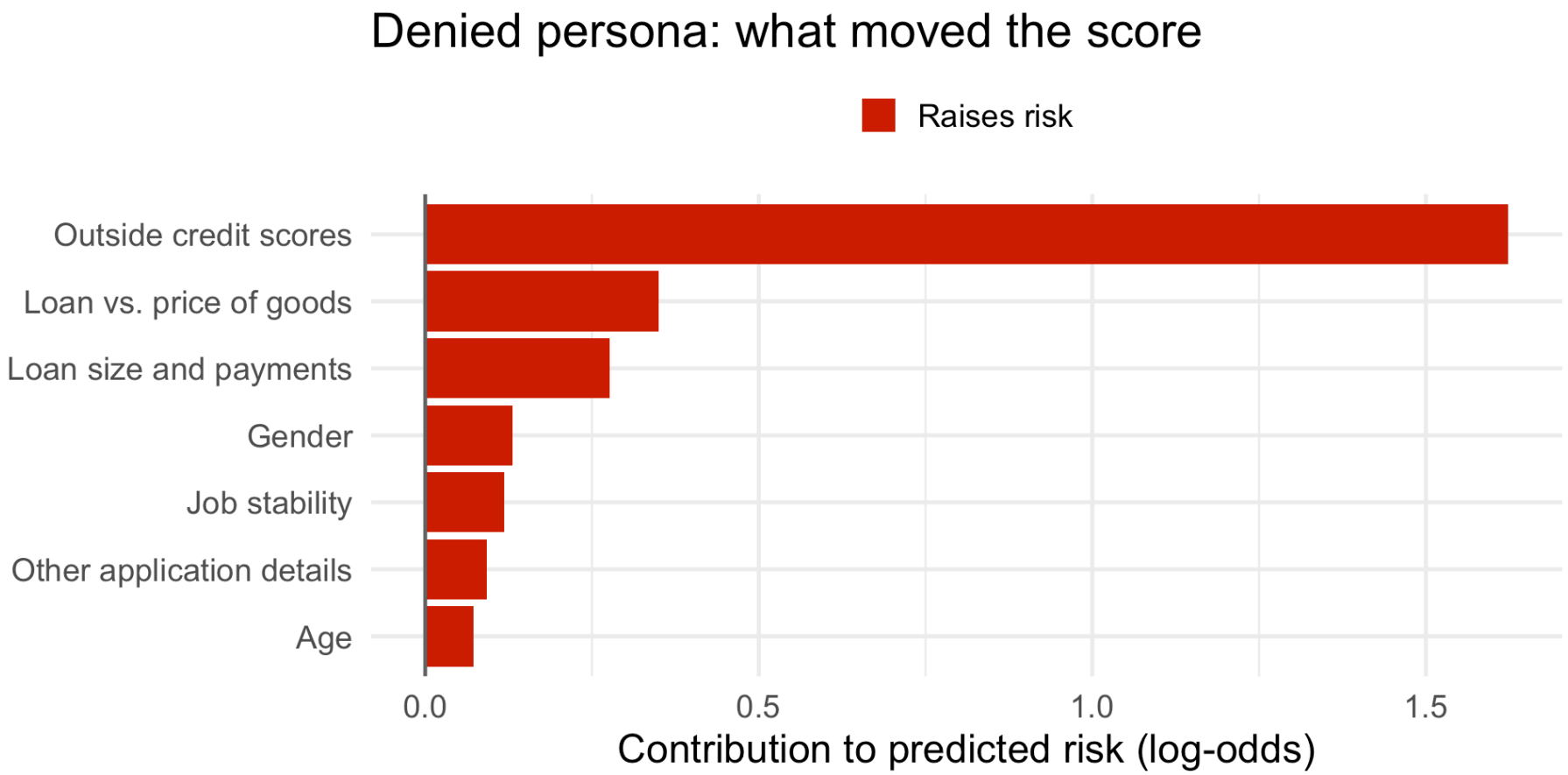
persona	column	value	contribution	direction
Approved	ext_source_mean	0.533	-0.252	lowers risk
Approved	NAME_EDUCATION_TYPEHigher education	1.000	-0.160	lowers risk
Approved	credit_to_annuity	23.603	-0.089	lowers risk
Approved	ltv	1.000	-0.089	lowers risk
Approved	CODE_GENDERM	0.000	-0.077	lowers risk
Approved	ext_source_max	0.579	0.067	raises risk
Approved	ext_source_min	0.487	-0.051	lowers risk
Approved	NAME_INCOME_TYPEState servant	1.000	-0.049	lowers risk
Denied	ext_source_mean	0.096	1.003	raises risk
Denied	ltv	1.422	0.356	raises risk
Denied	ext_source_min	0.063	0.273	raises risk
Denied	EXT_SOURCE_3	0.129	0.173	raises risk
Denied	AMT_ANNUITY	31261.500	0.139	raises risk
Denied	CODE_GENDERM	1.000	0.131	raises risk
Denied	credit_to_annuity	20.475	0.090	raises risk
Denied	employed_share_of_life	0.058	0.090	raises risk

```
write_csv(by_theme,   file.path(results_dir, "persona_drivers_by_theme.csv"))
write_csv(by_feature, file.path(results_dir, "persona_drivers_by_feature.csv"))
```

The charts below group the contributions into plain-English themes. Red bars pushed the predicted risk up; blue bars pushed it down. Contributions are on the log-odds scale, so they add up to the applicant's score.

```
for (p in c("Approved", "Denied")) {
  d <- by_theme |> filter(persona == p) |> slice_max(abs(contribution), n = 7)
  g <- ggplot(d, aes(contribution, fct_reorder(theme, abs(contribution)),
                     fill = contribution > 0)) +
    geom_col() +
    geom_vline(xintercept = 0, colour = "grey40") +
    scale_fill_manual(values = c(`TRUE` = "#CC0000", `FALSE` = "#1E2761"),
                      labels = c(`TRUE` = "Raises risk", `FALSE` = "Lowers risk"), name = NULL) +
    labs(title = paste0(p, " persona: what moved the score"),
         x = "Contribution to predicted risk (log-odds)", y = NULL) +
    theme_minimal(base_size = 18) + theme(legend.position = "top")
  print(g)
  ggsave(file.path(results_dir, paste0("persona_", tolower(p), "_drivers.png")),
         g, width = 11, height = 6, dpi = 200)
}
```


Approved persona: what moved the score
Lowers risk
Outside credit scores
Education
Loan vs. price of goods
Income
Loan size and payments
Gender
Housing and location
-0.3
-0.2
-0.1
0.0
Contribution to predicted risk (log-odds)



Age and gender in these explanations

Age and gender are protected characteristics in many lending contexts, and both appear in these explanations.

```
protected <- contrib |>
  filter(theme %in% c("Gender", "Age")) |>
  group_by(persona, theme) |>
  summarise(contribution = sum(contribution), .groups = "drop") |>
  mutate(direction = if_else(contribution > 0, "raises risk", "lowers risk"))

protected |>
  mutate(contribution = round(contribution, 3)) |>
  knitr::kable(caption = "Gender and age contributions (log-odds; positive = riskier)")
```

Gender and age contributions (log-odds; positive = riskier)

persona	theme	contribution	direction
Approved	Age	0.023	raises risk
Approved	Gender	-0.077	lowers risk
Denied	Age	0.073	raises risk
Denied	Gender	0.131	raises risk

```
pc <- function(p, t) protected$contribution[protected$persona == p & protected$theme == t]
```

Gender moved the denied applicant's score up by 0.13 and the approved applicant's down by 0.08 (log-odds). Age moved them by 0.07 and 0.02. These are small next to the outside credit scores and are not what drives either decision. But a model that scores applicants partly on gender and age would need a formal fairness and compliance review before any real use, including whether these features should be removed.

Summary for the slides

```
avg_risk <- mean(ho$prob)
fmt_profile <- function(r) {
  c(sprintf("- Predicted risk: %s (average applicant %s); risk rank %s of %s (1 = riskiest); decision: %s",
            pct(r$prob), pct(avg_risk), num(r$risk_rank), num(nrow(ho)), r$decision),
    sprintf("- Actual outcome: %s", if_else(r$TARGET == 1, "had payment difficulties", "repaid")),
    sprintf("- Age %d, %s, %s; employed %s years; %s",
            round(r$age_years), r$CODE_GENDER, r$NAME_FAMILY_STATUS,
            round(r$years_employed, 1), r$NAME_INCOME_TYPE),
    sprintf("- Occupation: %s; education: %s; housing: %s; children: %s",
            r$OCCUPATION_TYPE, r$NAME_EDUCATION_TYPE, r$NAME_HOUSING_TYPE, r$CNT_CHILDREN),
    sprintf("- Loan %s (%s) for goods priced %s, loan-to-value %.2f; income %s",
            num(r$AMT_CREDIT), r$NAME_CONTRACT_TYPE, num(r$AMT_GOODS_PRICE), r$ltv,
            num(r$AMT_INCOME_TOTAL)),
    sprintf("- Outside score average %.2f from %d score(s); credit bureau record: %s; blank fields: %d",
            r$ext_source_mean, as.integer(r$ext_source_count),
            if_else(r$has_bureau_record == 1, "yes", "no"), as.integer(r$fields_missing)))
}
lines <- c("# Persona results", "",
           sprintf("Rule: deny the riskiest %s (predicted risk at least %s).",
                   pct(deny_share, 0), pct(cutoff)),
           "",
           "These two applicants were intentionally selected as illustrative examples, not a representative
sample.",
           "Contributions are log-odds SHAP values: they show what the model's score is associated with, not what
causes default.",
           "")
# One bullet per theme, or a single "none" line when no theme moved risk that way.
driver_lines <- function(d) {
  if (nrow(d) == 0) return("  - None: no theme moved the predicted risk in this direction")
  paste0("  - ", d$theme, " (", round(d$contribution, 2), ")")
}
for (i in seq_len(nrow(personas))) {
  r <- personas[i, ]
  th <- by_theme |> filter(persona == r$persona)
  lower <- th |> filter(contribution < 0) |> head(3)
  raise <- th |> filter(contribution > 0) |> head(3)
```

```
  lines <- c(lines, paste0("## ", r$persona, " persona (holdout row ", r$row_id, ")"),
             fmt_profile(r),
             "- Biggest factors lowering risk:",
             driver_lines(lower),
             "- Biggest factors raising risk:",
             driver_lines(raise),
             sprintf("- Protected characteristics: gender %+.2f, age %+.2f (log-odds)",
                     pc(r$persona, "Gender"), pc(r$persona, "Age")),
             "")
}
writeLines(lines, file.path(results_dir, "persona_summary.md"))
cat(lines, sep = "\n")
```

```
## # Persona results
##
## Rule: deny the riskiest 10% (predicted risk at least 17.9%).
##
## These two applicants were intentionally selected as illustrative examples, not a representative sample.
## Contributions are log-odds SHAP values: they show what the model's score is associated with, not what causes
default.
##
## ## Approved persona (holdout row 46906)
## - Predicted risk: 3.6% (average applicant 8.0%); risk rank 42,130 of 61,502 (1 = riskiest); decision: Approve
## - Actual outcome: repaid
## - Age 29, F, Married; employed 4.3 years; State servant
## - Occupation: Core staff; education: Higher education; housing: House / apartment; children: 0
## - Loan 540,000 (Cash loans) for goods priced 540,000, loan-to-value 1.00; income 112,500
## - Outside score average 0.53 from 2 score(s); credit bureau record: no; blank fields: 8
## - Biggest factors lowering risk:
##   - Outside credit scores (-0.3)
##   - Education (-0.2)
##   - Loan vs. price of goods (-0.1)
## - Biggest factors raising risk:
##   - Other application details (0.05)
##   - Credit bureau history (0.03)
##   - Age (0.02)
## - Protected characteristics: gender -0.08, age +0.02 (log-odds)
##
## ## Denied persona (holdout row 22892)
## - Predicted risk: 60.7% (average applicant 8.0%); risk rank 1 of 61,502 (1 = riskiest); decision: Deny
## - Actual outcome: had payment difficulties
## - Age 39, M, Civil marriage; employed 2.3 years; Working
## - Occupation: Drivers; education: Secondary / secondary special; housing: House / apartment; children: 1
## - Loan 640,080 (Cash loans) for goods priced 450,000, loan-to-value 1.42; income 202,500
## - Outside score average 0.10 from 2 score(s); credit bureau record: yes; blank fields: 49
## - Biggest factors lowering risk:
##   - None: no theme moved the predicted risk in this direction
## - Biggest factors raising risk:
##   - Outside credit scores (1.62)
##   - Loan vs. price of goods (0.35)
```

```
##   - Loan size and payments (0.28)
## - Protected characteristics: gender +0.13, age +0.07 (log-odds)
```